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Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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Oil Viscosity, μ_o

1. Introduction

Viscosity is a fundamental property for describing the mobility of hydrocarbons through porous media. An oil with high viscosity exhibits greater resistance to flow, whereas an oil with low viscosity can move more easily through the pore space of the rock. Therefore, μ_o , has a direct influence on well productivity and reservoir performance.

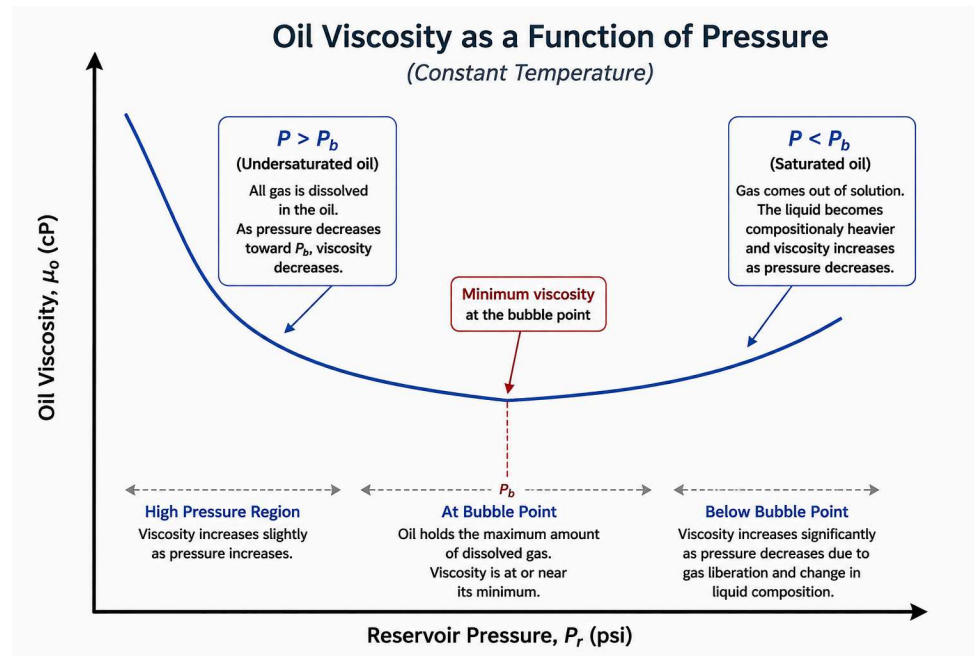
Oil viscosity primarily depends on pressure, temperature, fluid composition, and the amount of dissolved gas. In general:

- Temperature: An increase in temperature causes a decrease in oil viscosity. At higher temperatures, molecules have greater kinetic energy and can move more easily relative to one another.
- Composition: Oils containing a larger proportion of heavy hydrocarbon components exhibit higher viscosities. Consequently, heavy oils are generally considerably more viscous than light oils.
- Dissolved gas: The presence of dissolved gas significantly reduces oil viscosity. Light gas components contribute to reducing the viscosity of the liquid hydrocarbon phase.
- Pressure: The effect of pressure depends strongly on whether the oil is above or below its bubble-point pressure, p_b

$$\mu_o = f(y_i, T, R_s, p, \dots)$$

Oil volumes at surface or stock-tank conditions are conventionally reported at 60°F. Similarly, both stock-tank liquid volumes and surface gas volumes are expressed at standard conditions. The oil formation volume factor, B_o , is defined as the reservoir volume occupied by one stock tank barrel (STB) of oil, including the gas dissolved in solution, at reservoir pressure and temperature.

The following figure shows the typical variation of the oil formation volume factor with reservoir pressure for a black-oil reservoir fluid.



There are several numerical correlations available to compute the oil formation volume factor, B_o , some of the most widely used include:

- Beal
- Beggs
- Glaso
- Egbogad
- Kartootmodjo

2. Numerical solution

$\text{if } p \text{ equals } 1 \text{ atm and } T$

$\mu_o = 10^{x-1}$

$x = y \times T^{-1.163}$
 $y = 10^z$
 $z = 3.0324 - 0.02023 \times \gamma_{API}$

$p \leq p_b$

$\mu_o = a(\mu_o)^b$
 $a = 10.715(R_s + 100)^{-0.515}$
 $b = 5.44(R_s + 150)^{-0.338}$

$p > p_b$

$\mu_o = \mu_o \left(\frac{p}{p_b} \right)^m$
 $m = 2.6p^{1.187} \exp(-11.513 - 8.98 \times 10^{-5}p)$

2.1 Import libraries

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [2]: T=np.array([100,120,150])
gamma_o=31

gamma_g=0.9
gamma=1.2

pb=2200
Bob=1.3
co=5*10**-6

p=np.linspace(1000,3000,100)
```

2.3 Calculation and plotting

Note: To calculate the oil viscosity μ_{B_o} is required to know the solution gas-oil ratio $R_{s\$}$. Please review the corresponding notebook

```
In [9]: plt.figure(dpi=300, figsize=(8,4))

for i in range (len(T)):

    Rsb=gamma*((pb/18.2+1.4)*10**((0.0125*gamma_o-0.00091*T[i]))**1.2048)
    Rs_values=[]

    muo_values=[]

    for j in range (len(p)):
        if p[j]<pb:
            Rs=gamma*((p[j]/18.2+1.4)*10**((0.0125*gamma_o-0.00091*T[i]))**1.2048)
        else:
            Rs=Rsb

        Rs_values.append(Rs)

    # Properties at bubble point
    z=3.0324-0.02023*gamma_o
    y=10**z
    x=y*T[i]**-1.163
    muod=10**x-1

    a=10.715*(Rsb+100)**-0.515
    b=5.44*(Rsb+150)**-0.338
    muob=a*(muod)**b

    for j in range (len(p)):
        if p[j]<pb:

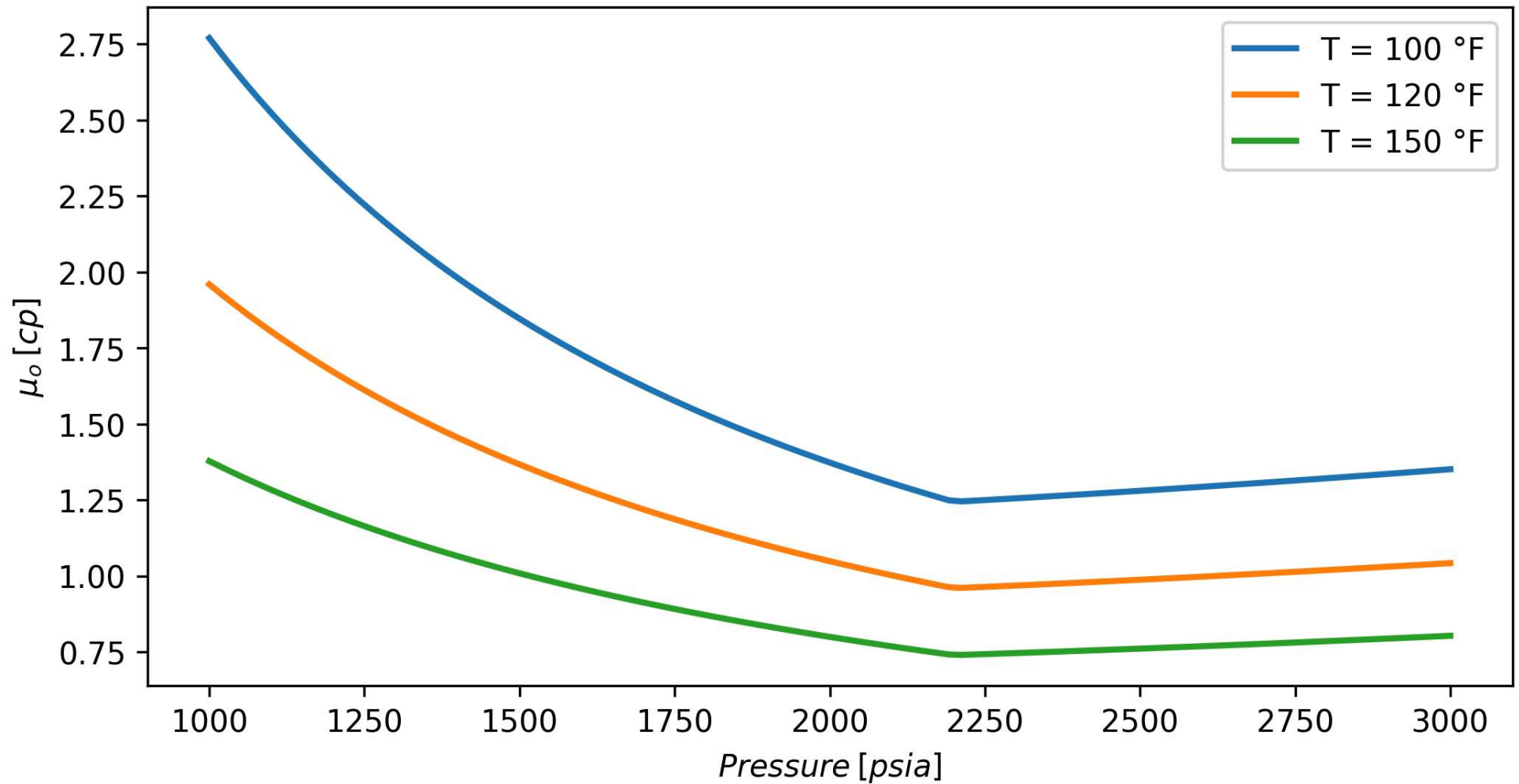
            a=10.715*(np.array(Rs_values[j])+100)**-0.515
            b=5.44*(np.array(Rs_values[j])+150)**-0.338
            muo=a*(muod)**b
        else:
            m=2.6*p[j]**1.187*np.exp(-11.513-8.98*10**(-5*p[j]))
            muo=muob*(p[j]/pb)**m

        muo_values.append(muo)

    plt.plot(p, muo_values, linestyle='-', linewidth=2,
             label=f'T = {T[i]} °F')

plt.xlabel('$Pressure\:[psia]$')
```

```
plt.ylabel('$\mu_o$ [cp]')  
# plt.grid(True)  
plt.legend()  
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In []: